OBJECTS OF THE SOURCE OF A MORPHISM OF SITES ARE COVERED BY IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The pullback of a morphism of sites is exact. Same chapter, the definition of a morphism of sites and the exactness of the pullback. Shows that the pullback of the final object is the final object.
- The left Kan extension as a colimit over a category of morphisms. Same chapter, the description of the left Kan extension as a colimit indexed by morphisms from the object into the images of objects. Turns the nonemptiness of a value into the existence of a morphism into the image of an object.

Source reference. Original section 00W3: sources/stacks-sites/pinned-original-source.tex, source lines 3113–3113. Navigation pointer invoked at proof line 3113; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $f : \mathcal{D} \to \mathcal{C}$ be a morphism of sites given by the functor $u : \mathcal{C} \to \mathcal{D}$. Given any object V of $\mathcal{D}$ there exists a covering $\{V_j \to V\}$ such that for every j there exists a morphism $V_j \to u(U_j)$ for some object U_j of $\mathcal{C}$.*

Proof. Since $f^{-1} = u_s$ is exact we have $f^{-1}* = *$ where $*$ denotes the final object of the category of sheaves (Example 00W3). Since $f^{-1}* = u_s*$ is the sheafification of u_p* we see there exists a covering $\{V_j \to V\}$ such that $(u_p*)(V_j)$ is nonempty.

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Since $(u_p*)(V_j)$ is a colimit over the category $\mathcal{I}^u_{V_j}$ whose objects are morphisms $V_j \to u(U)$ the lemma follows. $\square$